

■ POSTER E-COMMERCE PLATFORM

QA Test Execution Report

Report Date:	May 1-2, 2024
Project:	Poster E-Commerce Platform
Test Environment:	Staging
Execution Team:	QA Engineering Team
Total Test Cases:	20
Pass Rate:	90% (18/20)

Report Status: 2 Critical Issues Found - Action Required

■ EXECUTIVE SUMMARY

Test Execution Overview:

A comprehensive QA test execution was conducted for the Poster E-Commerce platform covering two sprints with a total of 20 test cases across 8 user stories. The test coverage includes smoke tests, functional tests, negative tests, performance tests, and edge case scenarios.

Key Results:

- Total Tests Executed: 20
- Tests Passed: 18 (90%)
- Tests Failed: 2 (10%)
- Critical Issues: 2
- Test Coverage: 100% of defined user stories

Overall Assessment: The platform demonstrates good functionality overall, but two critical issues must be resolved before production release: search functionality returning incomplete results and inadequate network error handling during checkout.

■ SPRINT 1: PRODUCT BROWSING & DETAILS

TC ID	Test Case Name	Type	Status	Comments
TC01	Display All Posters Homepage	Smoke	PASS ✓	Loaded in ~2 sec
TC01E1	No Posters Available State	Negative	PASS ✓	Empty state OK
TC01E2	Slow Loading Performance	Performance	PASS ✓	3.5 sec on 3G
TC02	Filter by Category	Functional	PASS ✓	Instant filtering
TC02E1	Empty Category Results	Negative	PASS ✓	Handled well
TC03	Valid Search Query	Functional	FAIL ✗	Only 3/10 results
TC03E1	Empty Search Query	Negative	PASS ✓	Validation OK
TC03E2	Special Characters Search	Negative	PASS ✓	Sanitized OK
TC04	Open Product Detail Page	Smoke	PASS ✓	1.2 sec load
TC05	Verify Product Details	Functional	PASS ✓	All fields OK
TC04E1	Missing Image Handling	Negative	PASS ✓	Placeholder shown

■ SPRINT 2: CART & CHECKOUT

TC ID	Test Case Name	Type	Status	Comments
TC07	Add Single Item to Cart	Smoke	PASS ✓	Toast shown
TC07E1	Add Duplicate Quantity	Functional	PASS ✓	Qty incremented
TC07E2	Add Out of Stock Item	Negative	PASS ✓	Button disabled
TC08	Update Item Quantity	Functional	PASS ✓	Total updated
TC09	Remove Item from Cart	Functional	PASS ✓	Instant removal
TC08E1	Quantity to Zero	Edge Case	PASS ✓	Item removed
TC09E1	Remove Last Item	Edge Case	PASS ✓	Empty state OK
TC10	Proceed to Checkout	Smoke	PASS ✓	0.8 sec load
TC11	Verify Order Summary	Functional	PASS ✓	Total correct
TC10E1	Empty Cart Checkout	Negative	PASS ✓	Redirected
TC10E2	Network Failure	Edge Case	FAIL ✗	Data lost

■ CRITICAL ISSUES IDENTIFIED

BUG-001: Search Functionality Returns Incomplete Results

Severity: HIGH | **Status:** OPEN | **Test Case:** TC03

Description: When searching for posters with keyword 'Nature', only 3 results returned instead of expected 10+. This indicates a critical issue with search indexing or query matching algorithm.

Impact: Users cannot effectively discover products, reducing conversion rates and customer satisfaction.

Root Cause: Likely database indexing issue or incomplete search implementation.

Recommended Fix: Review search query implementation and verify database indices are properly configured.

BUG-002: Network Error Handling During Checkout

Severity: HIGH | **Status:** OPEN | **Test Case:** TC10E2

Description: When network connectivity is lost during checkout, form data is not preserved. Users must restart the entire checkout process, resulting in form data loss.

Impact: Significantly degrades user experience and increases cart abandonment rates during network interruptions.

Root Cause: No client-side form data persistence or error recovery mechanism implemented.

Recommended Fix: Implement localStorage for temporary form data storage and add retry logic with data restoration.

■ RECOMMENDATIONS & ACTION ITEMS

IMMEDIATE ACTIONS (Critical - Before Release):

- 1. Fix search indexing issue (BUG-001) - impacts core product discovery functionality
- 2. Implement form data persistence for checkout (BUG-002) - enhances error resilience
- 3. Re-execute TC03 and TC10E2 after fixes are implemented

SHORT TERM (Next Sprint):

- 1. Add comprehensive error handling and recovery mechanisms for network scenarios
- 2. Implement localStorage/sessionStorage for temporary form state during checkout
- 3. Add automated test coverage for network resilience scenarios
- 4. Performance optimization targeting load times < 2 seconds across all pages
- 5. Implement logging and monitoring for search performance analytics

ONGOING/CONTINUOUS IMPROVEMENTS:

- 1. Establish CI/CD pipeline integration for automated regression testing
- 2. Conduct load testing for multiple concurrent users (10+, 50+, 100+ users)
- 3. Monitor search effectiveness with user analytics and search quality metrics
- 4. Expand test coverage to include integration tests and API contract testing
- 5. Implement continuous performance monitoring in production environment

■ DETAILED TEST METRICS

Metric	Value	Target	Status
Total Test Cases Executed	20	20	MET ✓
Tests Passed	18 (90%)	> 95%	BELOW TARGET
Tests Failed	2 (10%)	0	ACTION NEEDED
Critical Issues Found	2	0	ACTION NEEDED
Test Coverage by Sprint	100%	100%	MET ✓
Average Execution Time	1.8 hrs	< 2 hrs	MET ✓
Smoke Tests Completion	100%	100%	MET ✓
Negative Tests Coverage	100%	100%	MET ✓

■ CONCLUSION

The Poster E-Commerce platform demonstrates solid core functionality with a 90% test pass rate. The platform successfully handles most user workflows including product browsing, filtering, cart management, and checkout initiation. However, two critical issues must be addressed before production release:

1. **Search Functionality (BUG-001):** The search feature returns incomplete results, compromising the primary product discovery mechanism.
2. **Network Resilience (BUG-002):** The lack of error recovery during checkout creates a poor user experience when network connectivity is interrupted.

Recommendation: Delay production release until BUG-001 and BUG-002 are fixed and verified through regression testing. Estimated fix time: 3-5 business days. Once resolved, the platform will be ready for production deployment.

Next Steps: Development team should prioritize these issues immediately. QA team will conduct regression testing once fixes are implemented.

QA TEST EXECUTION REPORT

Generated: May 2, 2024 | Testing Team: QA Engineering | Environment: Staging

This report is confidential and intended for internal use by development and QA teams only.